5.3.6 Wake-up time from low-power modes

The wake-up times given in *Table 36* are the latency between the event and the execution of the first user instruction.

Table 36. Low-power mode wake-up times

Symbol	Parameter[1]	Conditions		Typ	Max	Unit
$t_{WUSLEEP}$	Wake-up time from Sleep to Run mode	HCLK = HSI48/4 = 12 MHz	Transiting to Run-mode execution in flash memory powered during Sleep mode	10	12	CPU clock cycles
			Transiting to Run-mode execution in flash memory not powered during Sleep mode	4.74	5.02	µs
$t_{WULPSTOP}$	Wake-up time from Stop mode	Clock after wake-up is HCLK = HSI48/4 = 12 MHz	Transiting to Run-mode execution in flash memory powered during Stop mode	2.7	3.1	µs
			Transiting to Run-mode execution in flash memory not powered during Stop mode	5.9	6.4	
			Transiting to Run-mode execution in SRAM	2.5	2.9	
t_{WUSTBY}	Wake-up time from Standby mode	Clock after wake-up is HCLK = HSI48/4 = 12 MHz	Transiting to Run mode	23.0	35.0	
t_{WUSHDN}	Wake-up time from Shutdown mode	Clock after wake-up is HCLK = HSI48/4 = 12 MHz	Transiting to Run mode	385.0	466.0	

1. Evaluated by characterization. Not tested in production.

5.3.7 External clock source characteristics

High-speed external user clock generated from an external source

In bypass mode the HSE oscillator is switched off and the input pin is a standard GPIO.

The external clock signal has to respect the I/O characteristics in *Section 5.3.13*. See *Figure 17* for recommended clock input waveform.

Table 37. High-speed external user clock characteristics

Symbol	Parameter[1]	Conditions	Min	Typ	Max	Unit
f_{HSE_ext}	User external clock source frequency	-	-	8	48	MHz
V_{HSEH}	Digital OSC_IN input pin high level voltage	-	$0.7 \times V_{DD}$	-	V_{DD}	V
V_{HSEL}	Digital OSC_IN input pin low level voltage	-	V_{SS}	-	$0.3 \times V_{DD}$	
$t_{w(HSEH)}$/ $t_{w(HSEL)}$	Digital OSC_IN high or low time	-	7	-	-	ns